

Part 3 - Manual Gradient Descent Calculations

Given model: $\hat{y} = X @ m + b$

Initial $m = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$, initial $b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$, $y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$, learning rate $\alpha = 0.001$, $n = 2$.

Cost function: $J(m,b) = (1/n) * \sum((\hat{y} - y)^2)$. Let error = $\hat{y} - y$.

Using the chain rule: $dJ/dm = (2/n) * X.T @ (\hat{y} - y)$, and $dJ/db = (2/n) * (\hat{y} - y)$.

Iteration 0 prediction:

$X @ m = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} @ \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 16 \end{bmatrix}$

$\hat{y} = \begin{bmatrix} 5 \\ 16 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$

error = $\hat{y} - y = \begin{bmatrix} 6 \\ 17 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$

$J = (1/2) * (1^2 + 11^2) = 61$

$dJ/dm = X.T @ \text{error} = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} @ \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$

$dJ/db = \text{error} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$

$m_{\text{new}} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} - 0.001 * \begin{bmatrix} 45 \\ 113 \end{bmatrix} = \begin{bmatrix} -1.045 \\ 1.887 \end{bmatrix}$

$b_{\text{new}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0.001 * \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} 0.999 \\ 0.989 \end{bmatrix}$

Iter	m1	m2	b1	b2	yhat1	yhat2	err1	err2	MSE
0	-1	2	1	1	6	17	1	11	61
1	-1.045	1.887	0.999	0.989	5.615	15.679	0.615	9.679	47.030633
2	-1.084331	1.788365	0.998385	0.979321	5.279149	14.525647	0.279149	8.525647	36.38229
3	-1.118713	1.702271	0.998106	0.970795	4.986206	13.518655	-0.013794	7.518655	28.265183
4	-1.148774	1.627126	0.99812	0.963277	4.730724	12.639442	-0.269276	6.639442	22.077347
5	-1.175062	1.561539	0.998389	0.956637	4.507945	11.871782	-0.492055	5.871782	17.359973

Observation: The MSE decreases after every update, so the parameters are moving in a direction that reduces the prediction error. The small learning rate makes the updates gradual and stable.